exceed the necessary payment, thereby requiring change). But everyone else carries only a denomination—a minimal "lump" of money—that exceeds the required payment. Therefore they cannot pay the landlord and hence rather than getting the infinite amount of money he expected, the landlord leaves with the paltry sum of $690.

Planck made use of a very similar strategy to reduce the ridiculous result of infinite energy in an oven to one that is finite. Here's how. Planck boldly guessed that the energy carried by an electromagnetic wave in the oven, like money, comes in lumps. The energy can be one times some fundamental "energy denomination," or two times it, or three times it, and so forth—but that's it. Just as you can't have one-third of a penny or two and a half quarters, Planck declared that when it comes to energy, no fractions are allowed. Now, our monetary denominations are determined by the United States Treasury. Seeking a more fundamental explanation, Planck suggested that the energy denomination of a wave—the minimal lump of energy that it can have—is determined by its frequency. Specifically, he posited that the minimum energy a wave can have is proportional to its frequency: larger frequency (shorter wavelength) implies larger minimum energy; smaller frequency (longer wavelength) implies smaller minimum energy. Roughly speaking, just as gentle ocean waves are long and luxurious while harsh ones are short and choppy, long-wavelength radiation is intrinsically less energetic than short-wavelength radiation.

Here's the punch line: Planck's calculations showed that this lumpiness of the allowed energy in each wave cured the previous ridiculous result of infinite total energy. It's not hard to see why. When an oven is heated to some chosen temperature, the calculations based on nineteenth-century thermodynamics predicted the common energy that each and every wave would supposedly contribute to the total. But like those comrades who cannot contribute the common amount of money they each owe the landlord because the monetary denomination they carry is too large, if the minimum energy a particular wave can carry exceeds the energy it is supposed to contribute, it can't contribute and instead lies dormant. Since, according to Planck, the minimum energy a wave can carry is proportional to its frequency, as we examine waves in the oven of ever larger frequency (shorter wavelength), sooner or later the minimum energy they can carry is